



Lab Task -03

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	
Clarity						
Content Quality						
Spelling & Grammar						
Organization and Formatting						
Total obtained mark						
Comments						

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Batch: 46th Section : B

Course Code: CSE124

Course Name: Object Oriented Programming Lab.

Course Teacher Name: Debabarata Mallick.

Designation: Lecturer

Submission Date: 15 /08/2026

17. Binary Addition

Write a Java program to add two binary numbers.

Code:

```
J labtask17.java 1
OOP Java Basic Programming Lab 3 > J labtask17.java > labtask17 > main(String[])
1  import java.util.Scanner;
2
3  public class labtask17 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input first binary number: ");
8          String bi1 = input.nextLine();
9
10         System.out.print(s: "Input second binary number: ");
11         String bi2 = input.nextLine();
12
13         int num1 = Integer.parseInt(bi1, radix: 2);
14         int num2 = Integer.parseInt(bi2, radix: 2);
15         int sum = num1 + num2;
16
17         String biSum = Integer.toBinaryString(sum);
18         System.out.println("Addition of two binary numbers: " + biSum);
19     }
20 }
21
```

Output:

```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labt
k17'
Input first binary number: 10
Input second binary number: 11
Addition of two binary numbers: 101
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation: I converted both binary numbers to decimal, added them and converted the answer back to binary.

18. Binary Multiplication

Write a Java program to multiply two binary numbers.

Code:

```
labtask17.java 1  BinaryMultiplication.java 1  labtask18.java 1 X
OOP Java Basic Programming Lab 3 > labtask18.java > ...

1  import java.util.Scanner;
2
3  public class labtask18 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input the first binary number: ");
8          String bi1 = input.nextLine();
9
10         System.out.print(s: "Input the second binary number: ");
11         String bi2 = input.nextLine();
12
13         int num1 = Integer.parseInt(bi1, radix: 2);
14         int num2 = Integer.parseInt(bi2, radix: 2);
15         int multiple = num1 * num2;
16
17         String bimult = Integer.toBinaryString(multiple);
18         System.out.println("Multiplication of two binary numbers: " + bimult);
19     }
20 }
21
```

Output:

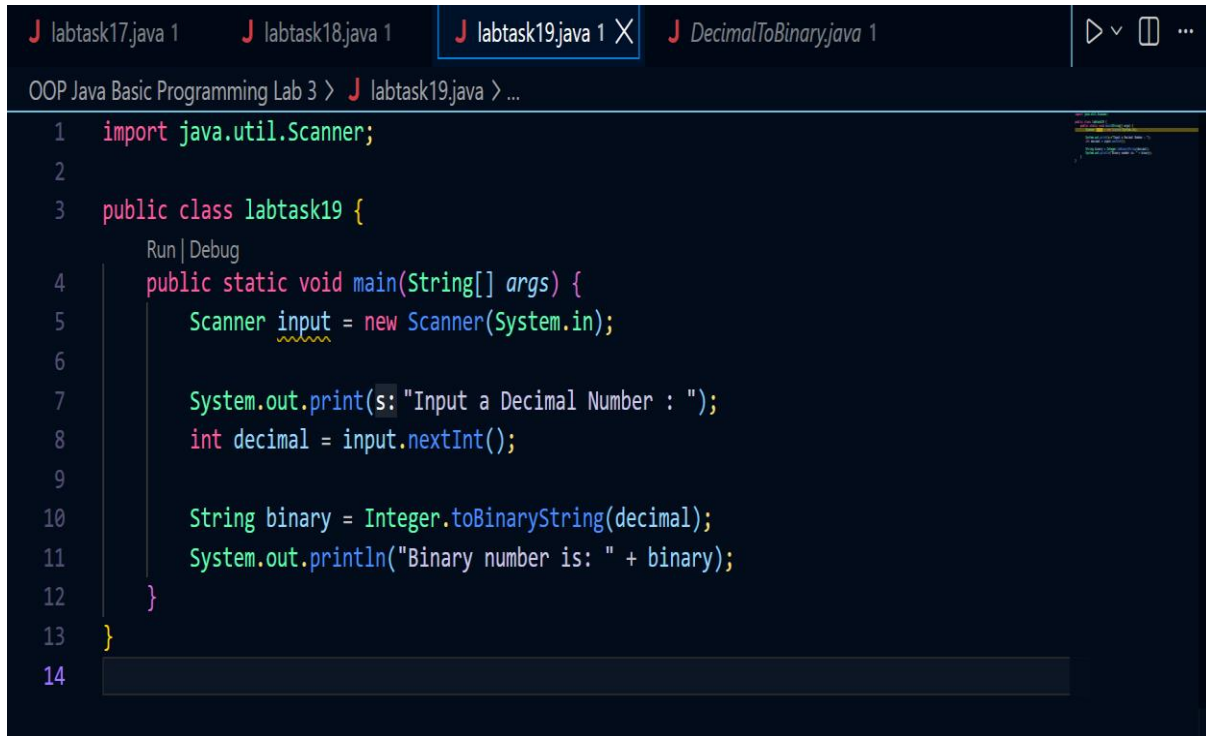
```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k18'
Input the first binary number: 10
Input the second binary number: 11
Multiplication of two binary numbers: 110
```

Explanation: I converted to decimal, multiplied them, and flipped the result back to binary.

19. Decimal to Binary Converter

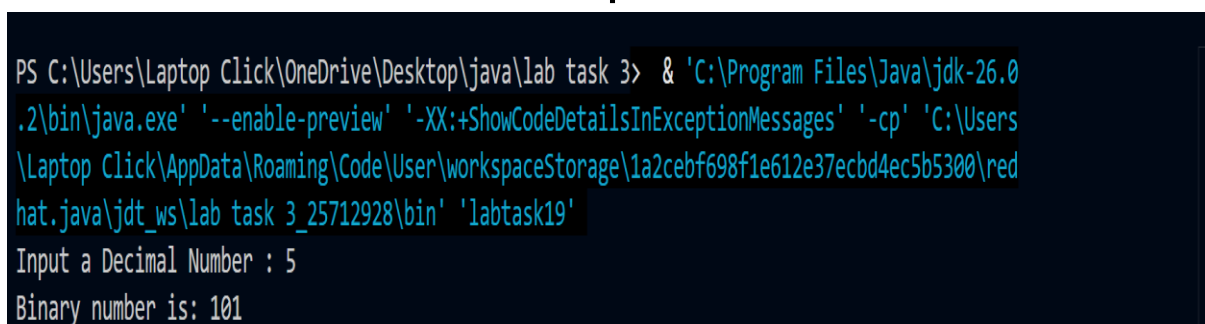
Write a Java program to convert an integer number to a binary number.

Code:

A screenshot of an IDE window with a dark theme. The title bar shows several tabs: 'labtask17.java 1', 'labtask18.java 1', 'labtask19.java 1' (which is the active tab), and 'DecimalToBinary.java 1'. The breadcrumb navigation shows 'OOP Java Basic Programming Lab 3 > labtask19.java > ...'. The code editor displays the following Java code:

```
1 import java.util.Scanner;  
2  
3 public class labtask19 {  
4     public static void main(String[] args) {  
5         Scanner input = new Scanner(System.in);  
6  
7         System.out.print(s: "Input a Decimal Number : ");  
8         int decimal = input.nextInt();  
9  
10        String binary = Integer.toBinaryString(decimal);  
11        System.out.println("Binary number is: " + binary);  
12    }  
13 }  
14
```

Output:

A screenshot of a terminal window with a dark background. It shows the command to run the Java program and its output:

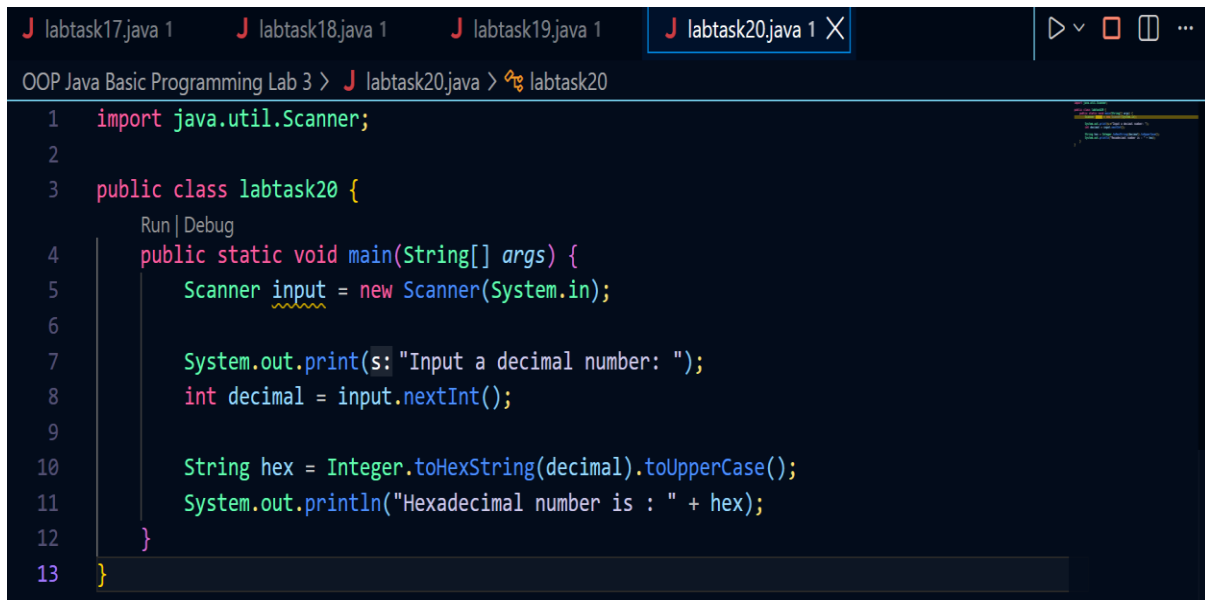
```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0  
.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users  
\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\red  
hat.java\jdt_ws\lab task 3_25712928\bin' 'labtask19'  
Input a Decimal Number : 5  
Binary number is: 101
```

Explanation: Java already has a built-in method (toBinaryString) for this, so I just called it and printed the output.

20. Decimal to Hexadecimal Converter

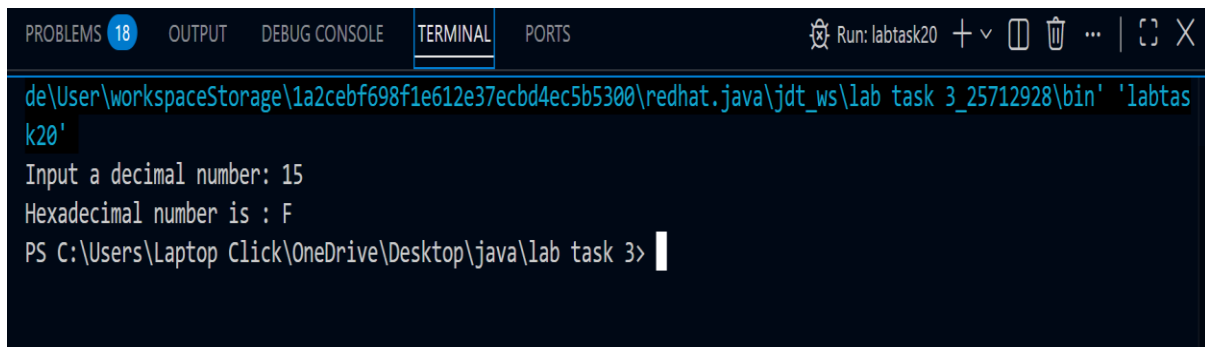
Write a Java program to convert a decimal number to a hexadecimal number.

Code:



```
1  import java.util.Scanner;
2
3  public class labtask20 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input a decimal number: ");
8          int decimal = input.nextInt();
9
10         String hex = Integer.toHexString(decimal).toUpperCase();
11         System.out.println("Hexadecimal number is : " + hex);
12     }
13 }
```

Output:



```
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k20'
Input a decimal number: 15
Hexadecimal number is : F
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation: Converts decimal to hex. The only catch: Java outputs lowercase letters by default (like f), so I used `.toUpperCase()` to get standard uppercase letters like F

21. Decimal to Octal Converter

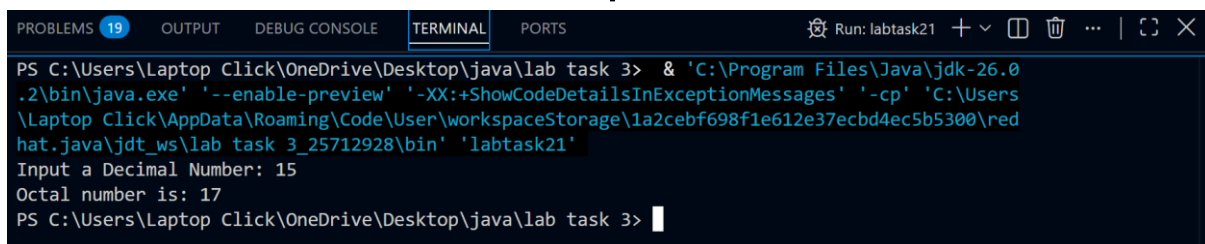
Write a Java program to convert a decimal number to an octal number.

Code:



```
labtask17.java 1    J labtask18.java 1    J labtask19.java 1    J labtask20.java 1    J labtask21.java 1 X  ▸ ▾ □ ▢ ...
OOP Java Basic Programming Lab 3 > J labtask21.java > ...
1  import java.util.Scanner;
2
3  public class labtask21 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner input = new Scanner(System.in);
7
8          System.out.print(s: "Input a Decimal Number: ");
9          int decimal = input.nextInt();
10
11         String octal = Integer.toOctalString(decimal);
12         System.out.println("Octal number is: " + octal);
13     }
14 }
```

Output:



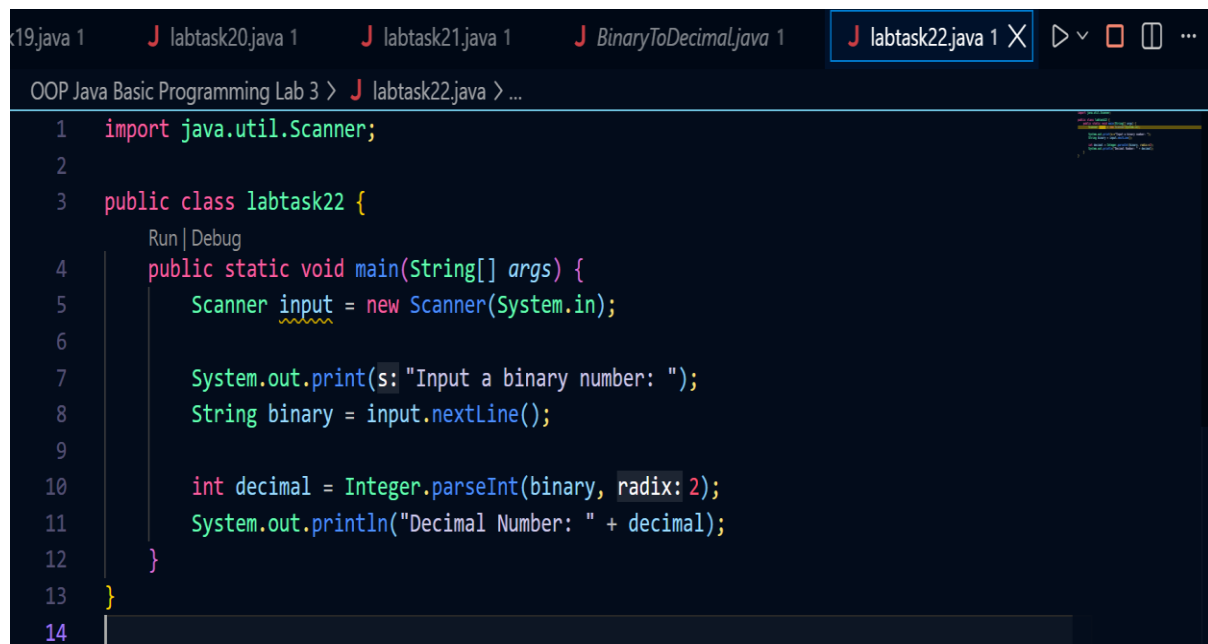
```
PROBLEMS 19  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0
.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\red
hat.java\jdt_ws\lab task 3_25712928\bin' 'labtask21'
Input a Decimal Number: 15
Octal number is: 17
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> |
```

Explanation: Same concept as binary and hex. Java has an octal method.

22. Binary to Decimal Converter

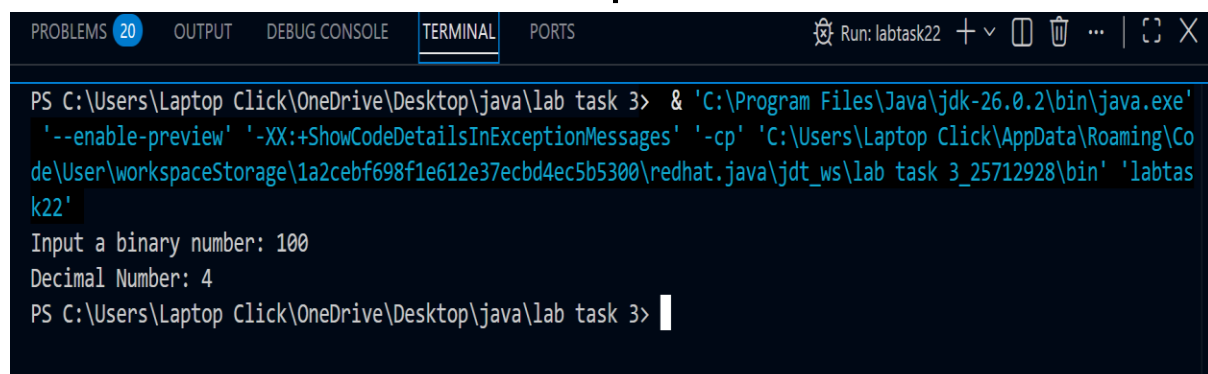
Write a Java program to convert a binary number to a decimal number.

Code:



```
1  import java.util.Scanner;
2
3  public class labtask22 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input a binary number: ");
8          String binary = input.nextLine();
9
10         int decimal = Integer.parseInt(binary, radix: 2);
11         System.out.println("Decimal Number: " + decimal);
12     }
13 }
14
```

Output:



```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k22'
Input a binary number: 100
Decimal Number: 4
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation : I used `parseInt` and specified base 2, letting Java handle the heavy lifting.

23. Binary to Hexadecimal Converter

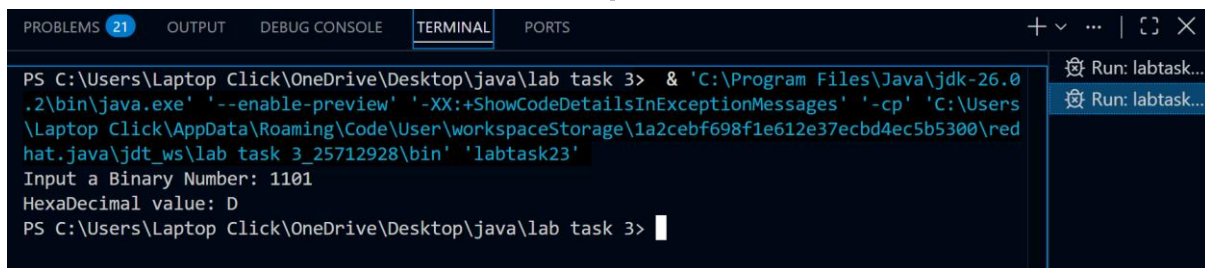
Write a Java program to convert a binary number to a hexadecimal number.

Code:



```
1 import java.util.Scanner;
2
3 public class labtask23 {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6
7         System.out.print(s: "Input a Binary Number: ");
8         String binary = input.nextLine();
9
10        int decimal = Integer.parseInt(binary, radix: 2);
11        String hex = Integer.toHexString(decimal).toUpperCase();
12        System.out.println("HexaDecimal value: " + hex);
13    }
14 }
15
```

Output:



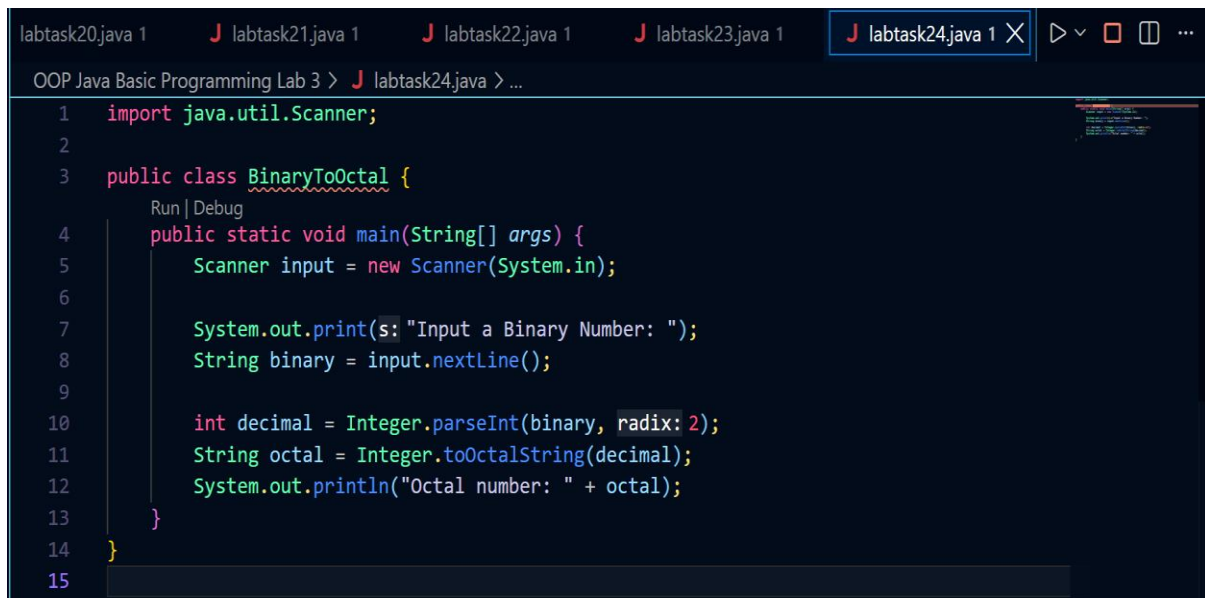
```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\red hat.java\jdt_ws\lab task 3_25712928\bin' 'labtask23'
Input a Binary Number: 1101
HexaDecimal value: D
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation : Converting straight from binary to hex felt a bit messy. Breaking it into two small steps binary to decimal, then decimal to hex . made it way simpler.

24. Binary to Octal Converter

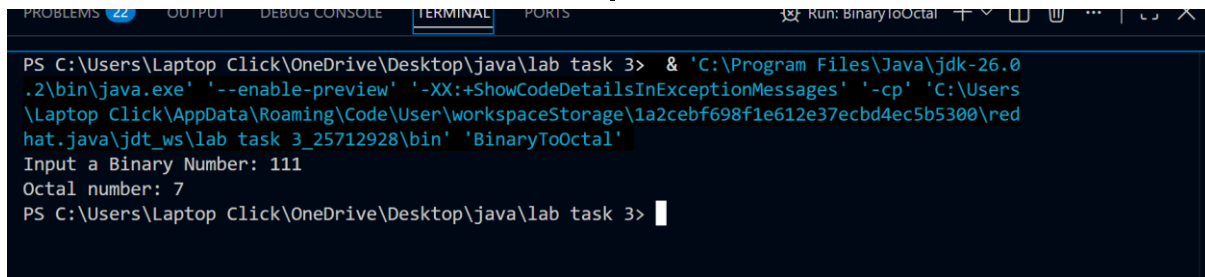
Write a Java program to convert a binary number to an octal number.

Code:



```
labtask20.java 1 labtask21.java 1 labtask22.java 1 labtask23.java 1 labtask24.java 1 X
OOP Java Basic Programming Lab 3 > labtask24.java > ...
1 import java.util.Scanner;
2
3 public class BinaryToOctal {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6
7         System.out.print("Input a Binary Number: ");
8         String binary = input.nextLine();
9
10        int decimal = Integer.parseInt(binary, radix: 2);
11        String octal = Integer.toOctalString(decimal);
12        System.out.println("Octal number: " + octal);
13    }
14 }
15
```

Output:



```
PROBLEMS 22 OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: BinaryToOctal
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0
.2\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users
\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\red
hat.java\jdt_ws\lab task 3_25712928\bin' 'BinaryToOctal'
Input a Binary Number: 111
Octal number: 7
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation : Reused the same two-step trick: binary to decimal first, then decimal to octal.

25. Octal to Decimal Converter

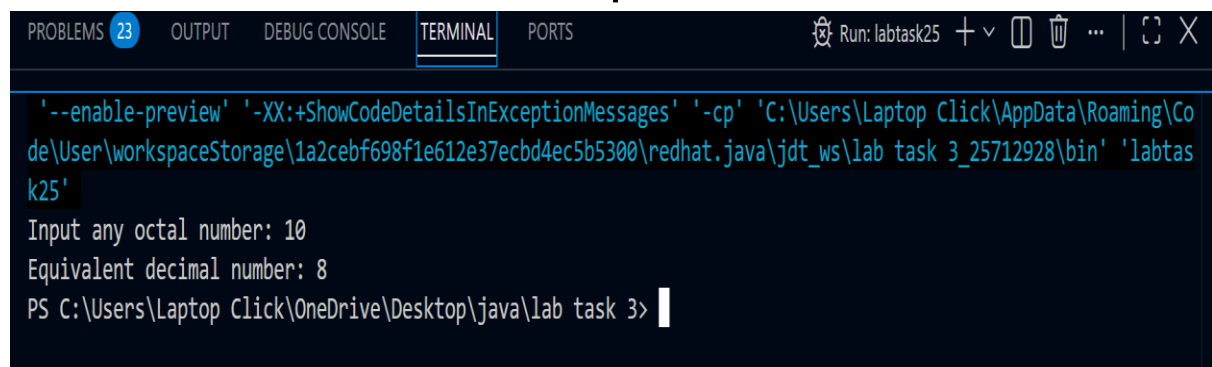
Write a Java program to convert a octal number to a decimal number.

Code:



```
1  import java.util.Scanner;
2
3  public class labtask25 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input any octal number: ");
8          String octal = input.nextLine();
9
10         int decimal = Integer.parseInt(octal, radix: 8);
11         System.out.println("Equivalent decimal number: " + decimal);
12     }
13 }
14
```

Output:



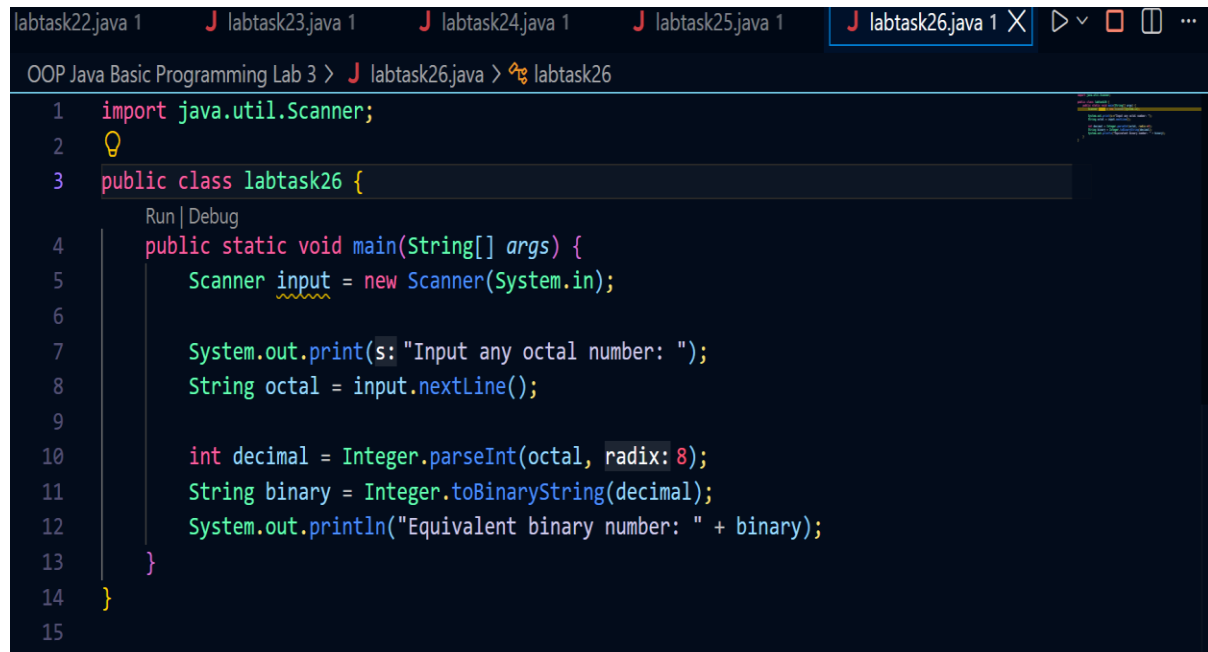
```
PROBLEMS 23 OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: labtask25 + v [ ] [ ] ... [ ] [X]
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtask25'
Input any octal number: 10
Equivalent decimal number: 8
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation : Converts base-8 back to base-10. I just passed base 8 into parseInt, and Java worked out the value.

26. Octal to Binary Converter

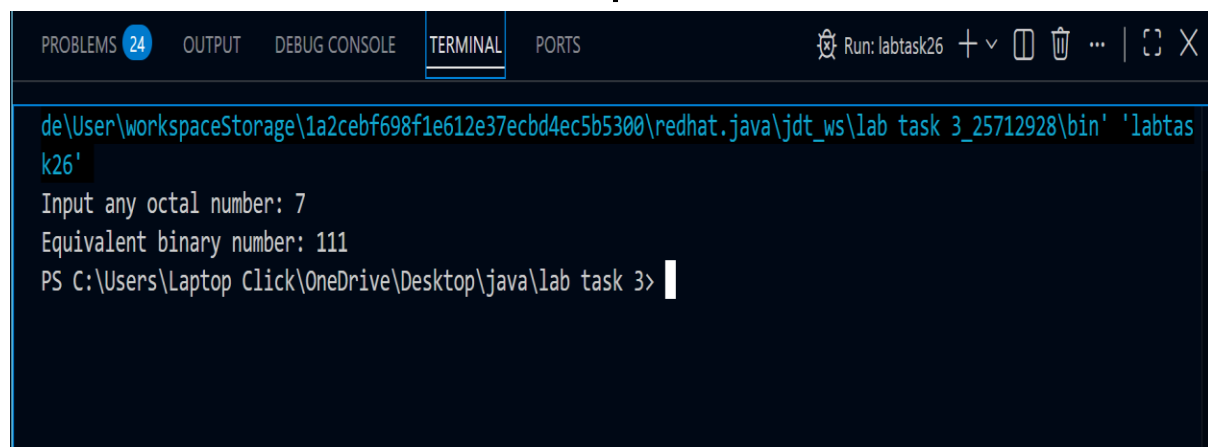
Write a Java program to convert a octal number to a binary number.

Code:



```
labtask22.java 1 labtask23.java 1 labtask24.java 1 labtask25.java 1 labtask26.java 1 X
OOP Java Basic Programming Lab 3 > labtask26.java > labtask26
1 import java.util.Scanner;
2
3 public class labtask26 {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6
7         System.out.print(s: "Input any octal number: ");
8         String octal = input.nextLine();
9
10        int decimal = Integer.parseInt(octal, radix: 8);
11        String binary = Integer.toBinaryString(decimal);
12        System.out.println("Equivalent binary number: " + binary);
13    }
14 }
15
```

Output:



```
PROBLEMS 24 OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: labtask26 + v [ ] [ ] ... | [ ] X
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k26'
Input any octal number: 7
Equivalent binary number: 111
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation : Octal to decimal, then decimal to binary. By this task, the two-step approach felt like second nature.

27. Octal to Hexadecimal Converter

Write a Java program to convert a octal number to a hexadecimal number.

Code:

```
labtask27.java 1 X
OOP Java Basic Programming Lab 3 > labtask27.java > labtask27 > main(String[])
1  import java.util.Scanner;
2
3  public class labtask27 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input a octal number : ");
8          String octal = input.nextLine();
9
10         int decimal = Integer.parseInt(octal, radix: 8);
11         String hex = Integer.toHexString(decimal).toUpperCase();
12         System.out.println("Hexadecimal number: " + hex);
13     }
14 }
```

Output:

```
PROBLEMS 25 OUTPUT DEBUG CONSOLE TERMINAL PORTS Run: labtask27 + v [ ] [ ] ... | [ ] [ ] X
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'lbtas
k27'
Input a octal number : 100
Hexadecimal number: 40
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> |
```

Explanation octal-decimal-hex. It's essentially the same couple lines of code, just tweaking the starting and ending bases.

28. Hexadecimal to Decimal Converter

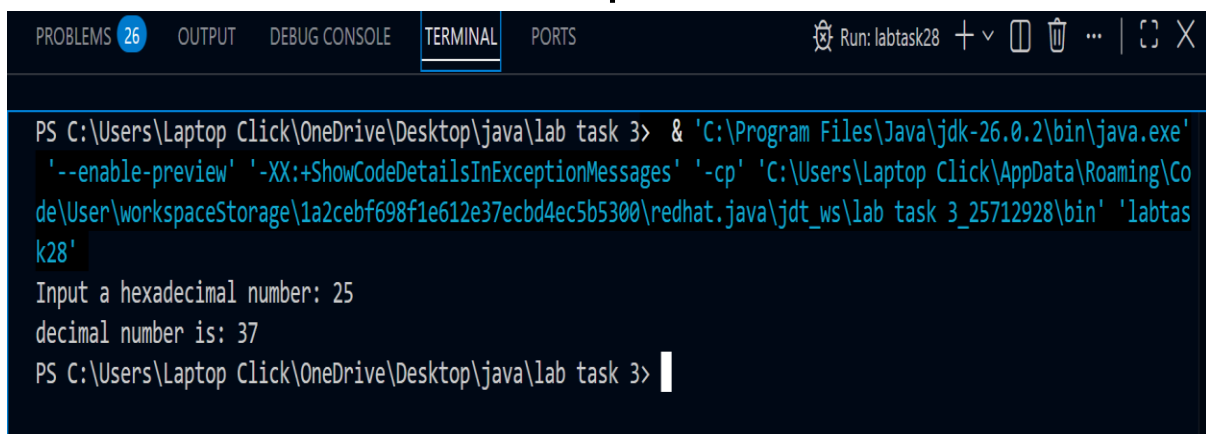
Write a Java program to convert a hexadecimal value into a decimal number.

Code:



```
1 import java.util.Scanner;
2
3 public class labtask28 {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6
7         System.out.print(s: "Input a hexadecimal number: ");
8         String hex = input.nextLine();
9
10        int decimal = Integer.parseInt(hex, radix: 16);
11        System.out.println("decimal number is: " + decimal);
12    }
13 }
```

Output:



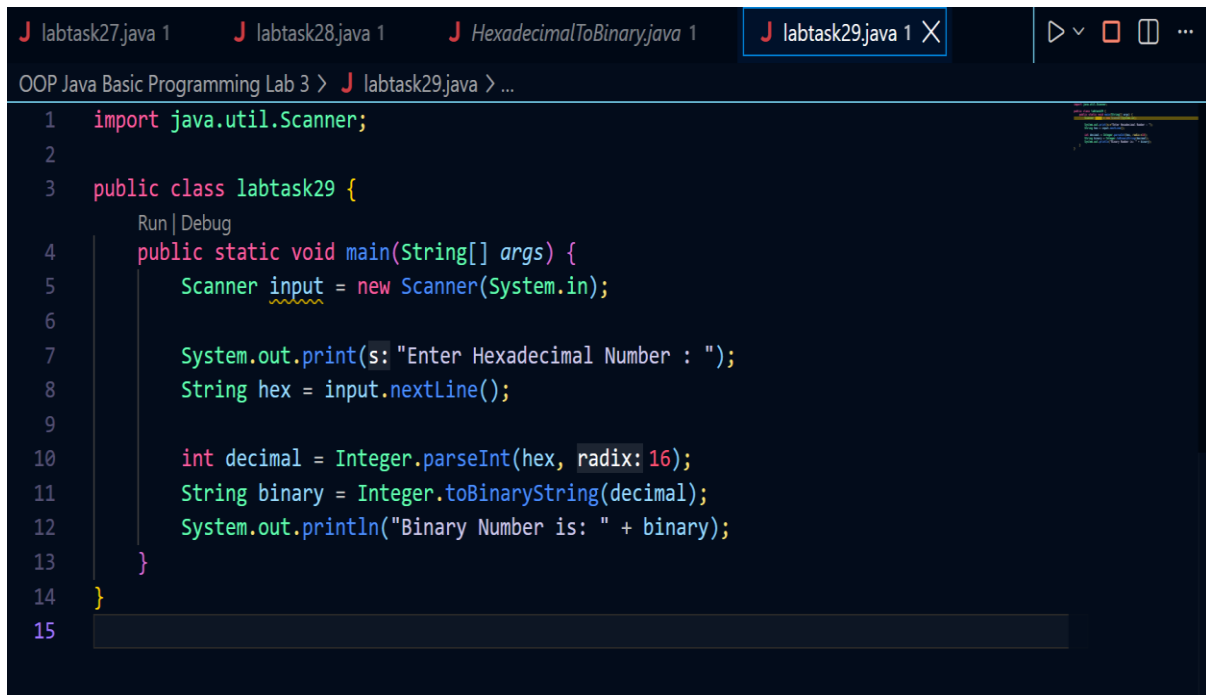
```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k28'
Input a hexadecimal number: 25
decimal number is: 37
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> |
```

Explanation: Converts hex back to decimal. Java automatically understands that letters like A through F represent values when you tell it to use base 16.

29. Hexadecimal to Binary Converter

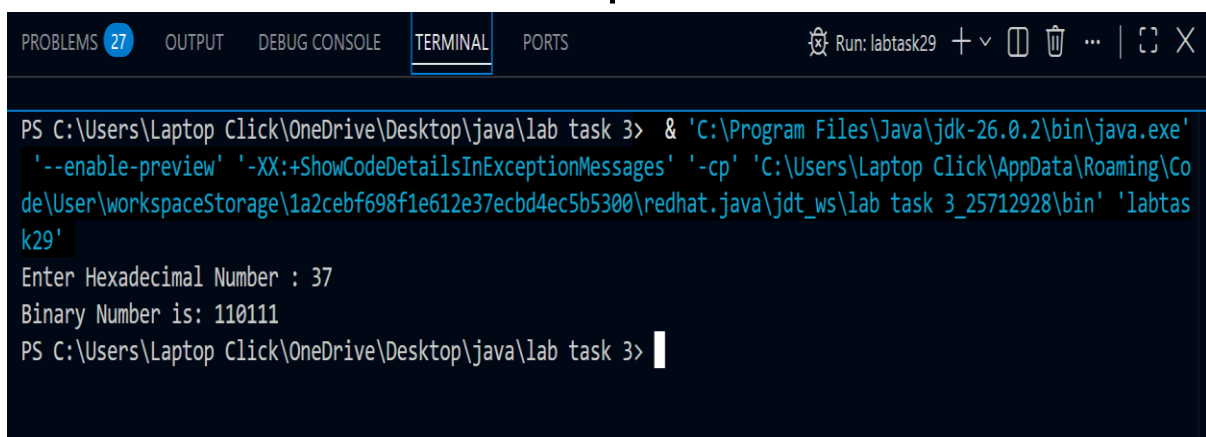
Write a Java program to convert a hexadecimal number into a binary number.

Code:



```
1 import java.util.Scanner;
2
3 public class labtask29 {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6
7         System.out.print(s: "Enter Hexadecimal Number : ");
8         String hex = input.nextLine();
9
10        int decimal = Integer.parseInt(hex, radix: 16);
11        String binary = Integer.toBinaryString(decimal);
12        System.out.println("Binary Number is: " + binary);
13    }
14 }
15
```

Output:



```
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3> & 'C:\Program Files\Java\jdk-26.0.2\bin\java.exe'
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Co
de\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtas
k29'
Enter Hexadecimal Number : 37
Binary Number is: 110111
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation: Hex to decimal, then decimal to binary.

30. Hexadecimal to Octal Converter

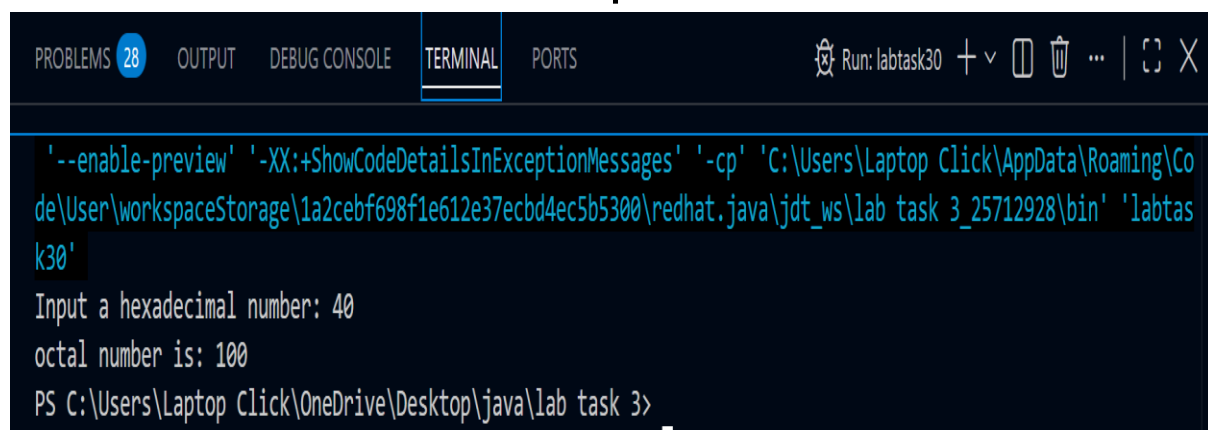
Write a Java program to convert a hexadecimal value into an octal number.

Code:



```
1  import java.util.Scanner;
2
3  public class labtask30 {
4      public static void main(String[] args) {
5          Scanner input = new Scanner(System.in);
6
7          System.out.print(s: "Input a hexadecimal number: ");
8          String hex = input.nextLine();
9
10         int decimal = Integer.parseInt(hex, radix: 16);
11         String octal = Integer.toOctalString(decimal);
12         System.out.println("octal number is: " + octal);
13     }
14 }
15
```

Output:



```
PROBLEMS 28 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Run: labtask30
'--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Laptop Click\AppData\Roaming\Code\User\workspaceStorage\1a2cebf698f1e612e37ecbd4ec5b5300\redhat.java\jdt_ws\lab task 3_25712928\bin' 'labtask30'
Input a hexadecimal number: 40
octal number is: 100
PS C:\Users\Laptop Click\OneDrive\Desktop\java\lab task 3>
```

Explanation: hex to octal using decimal as the bridge!